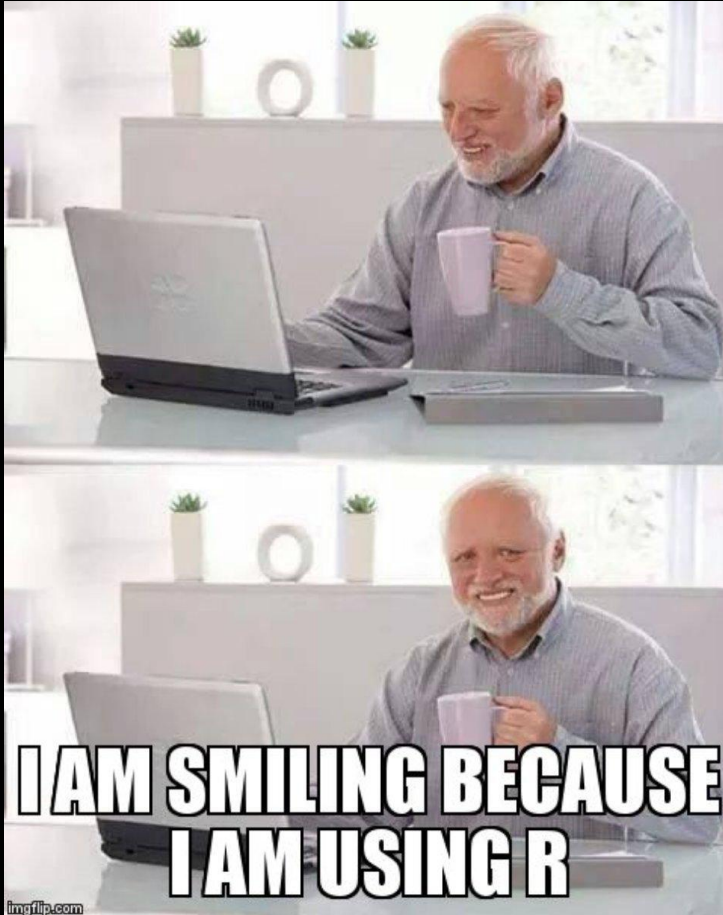




MES 503 Lab 2

Kayla Blincow, PhD

Announcements

- I misspoke on Monday about alpha and p values

Organizing Ourselves

- File Structure
- File Naming
- Documentation/Comments
- Why we must do this
 - Publishing things takes a long time



BEING ORGANIZED

is for people who are too lazy to look for their stuff.

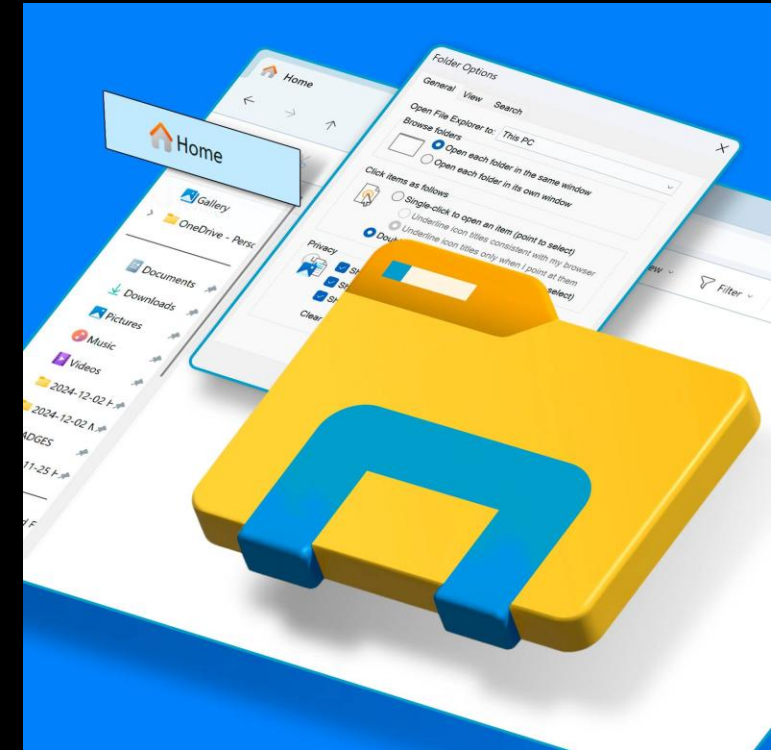
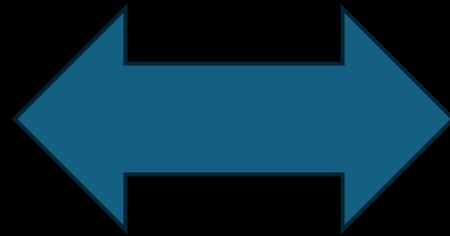
[VERY DEMOTIVATIONAL .com](http://VERYDEMOTIVATIONAL.com)

GOAL: Future you should be grateful for past you (not angry with past you)

How R integrates with your computer files

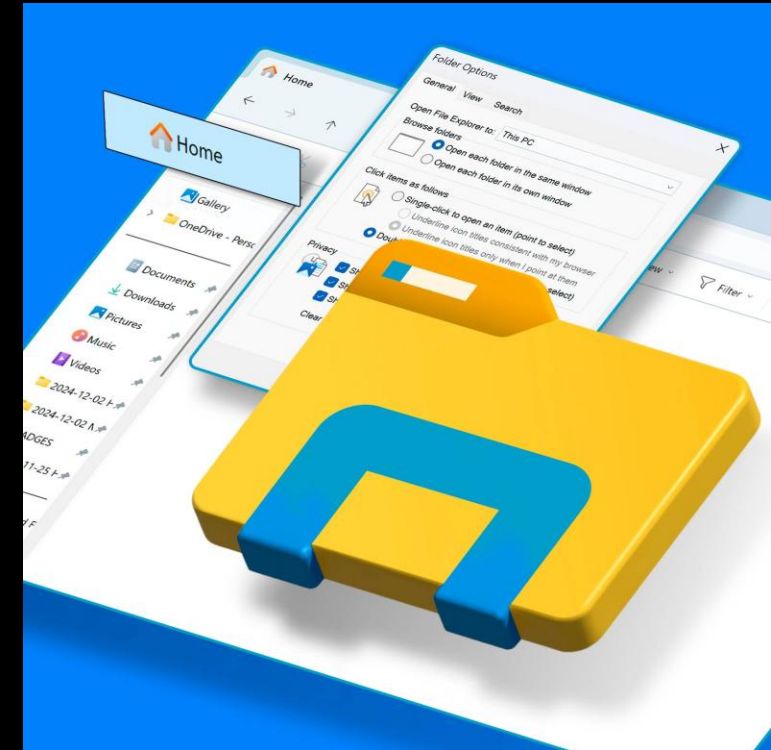
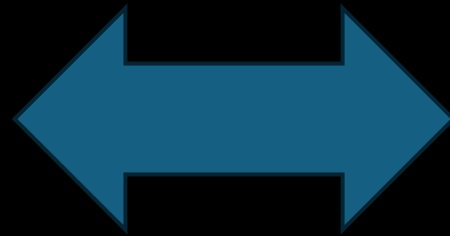


You create/edit/import/export files in R: e.g. .csv, .R, .rds, etc.



When you hit save, they automatically get stored in your **working directory**, which is a file on your computer

How R integrates with your computer files

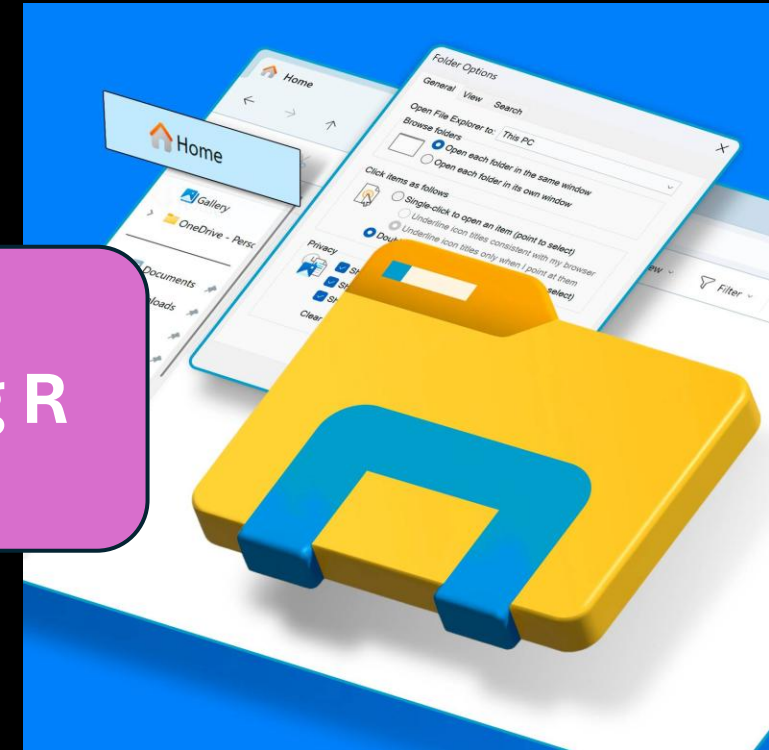


We need to set up a file structure on our computers that makes sense and then tell R our working directory is within that file structure.

How R integrates with your computer files



We will do this
second part using R
Projects.



We need to set up a file structure on our computers that makes sense and **then tell R our working directory is within that file structure.**

Let's set ourselves up for all our lab things!

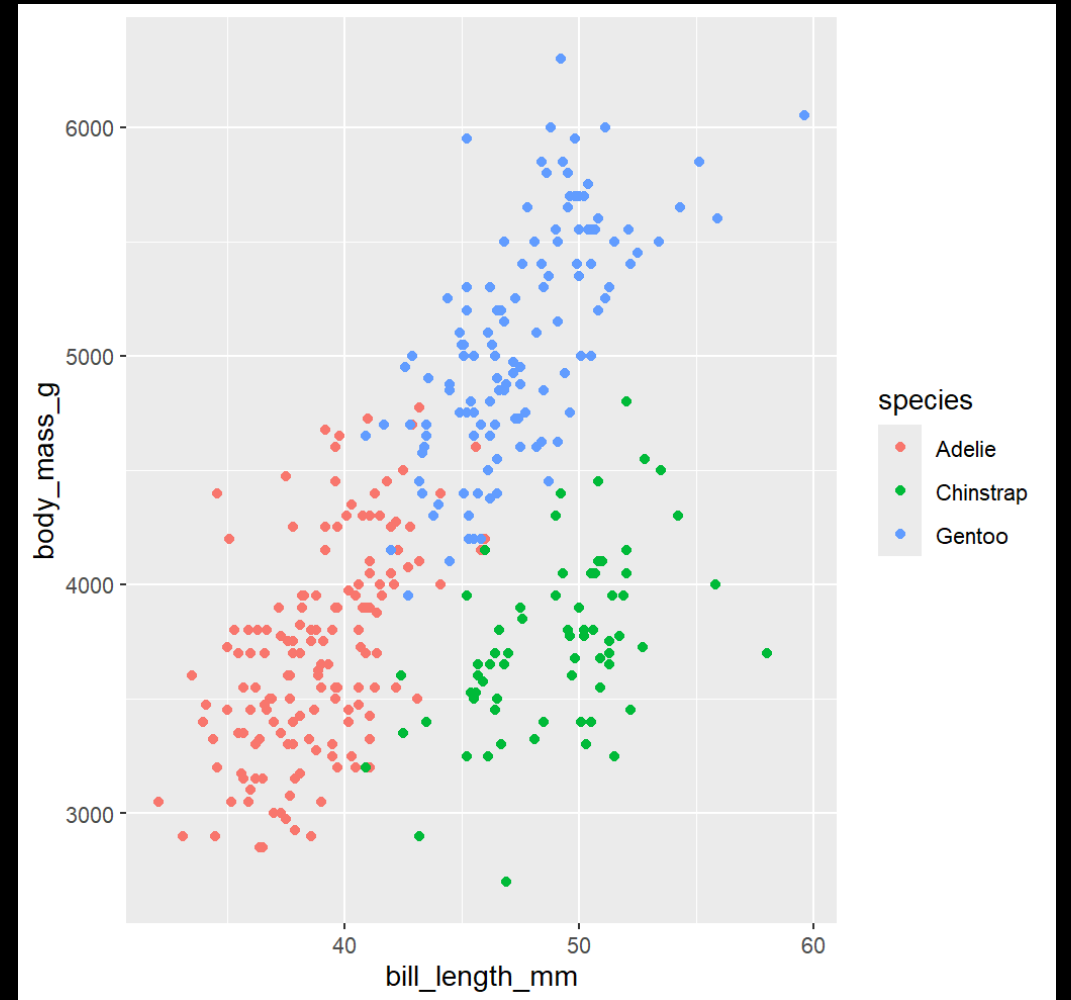
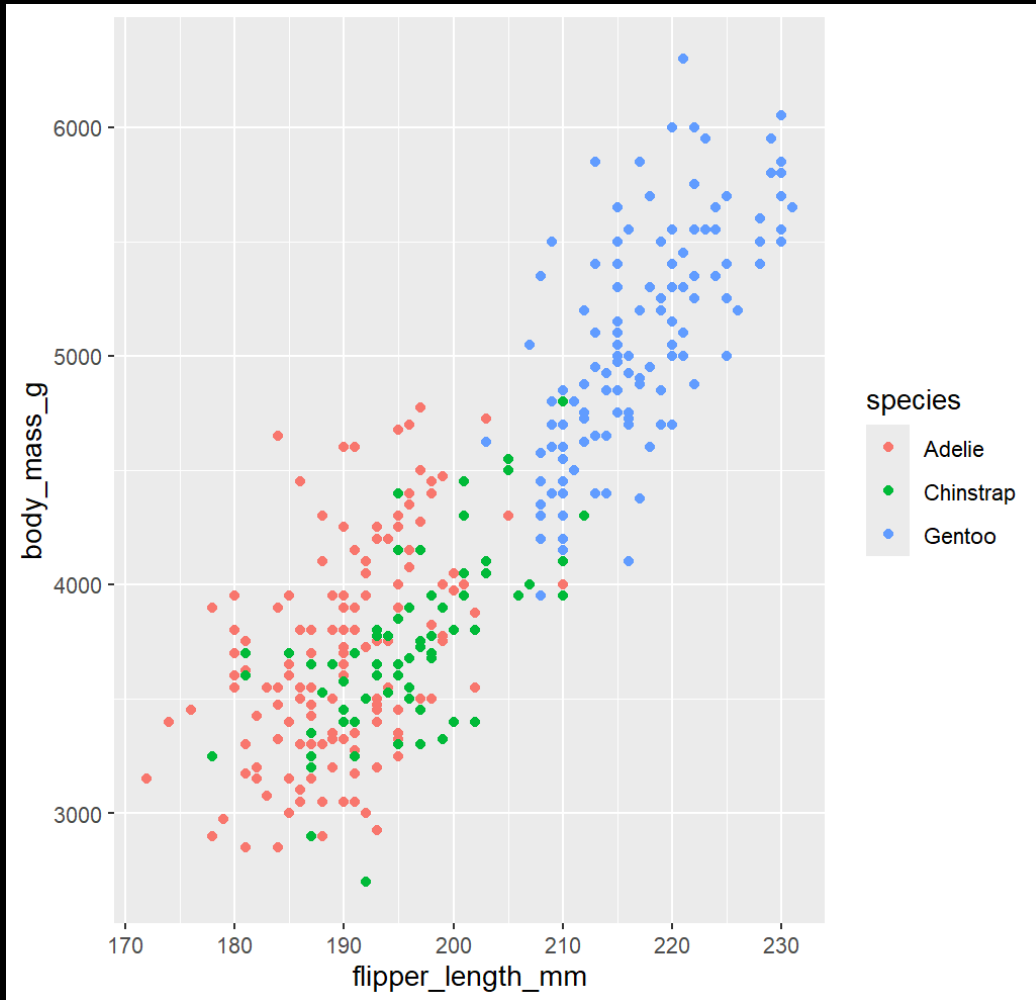
1. Create a folder on your computer to store all your lab materials
 - Name it something meaningful (e.g. MES503Lab)
2. Create an R Project within that folder
 - This will automatically set this parent folder as your working directory
3. Create a folder for this week's materials (e.g. Week 2)
4. Download this week's materials from Brightspace and store them in your Week 2 folder

Switch to ze code!

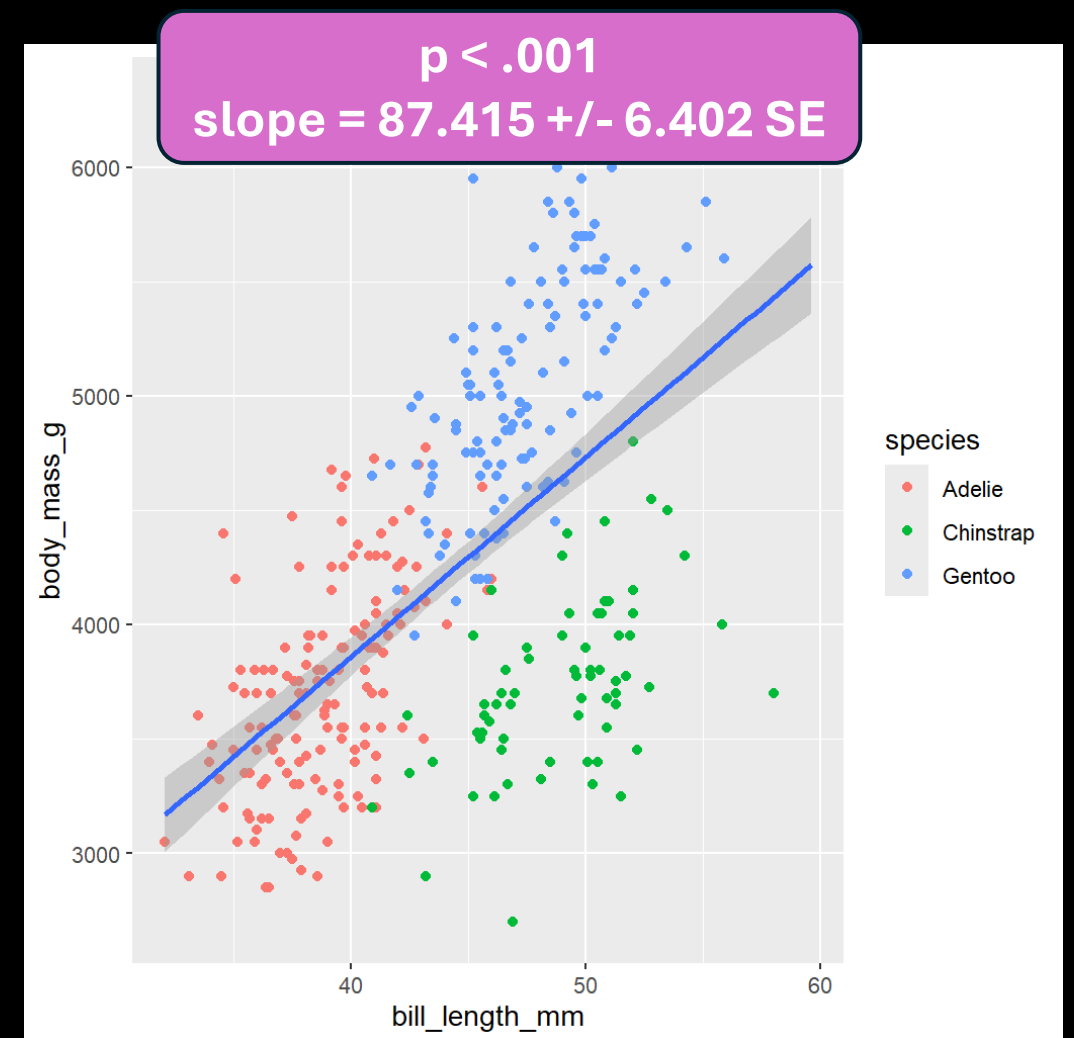
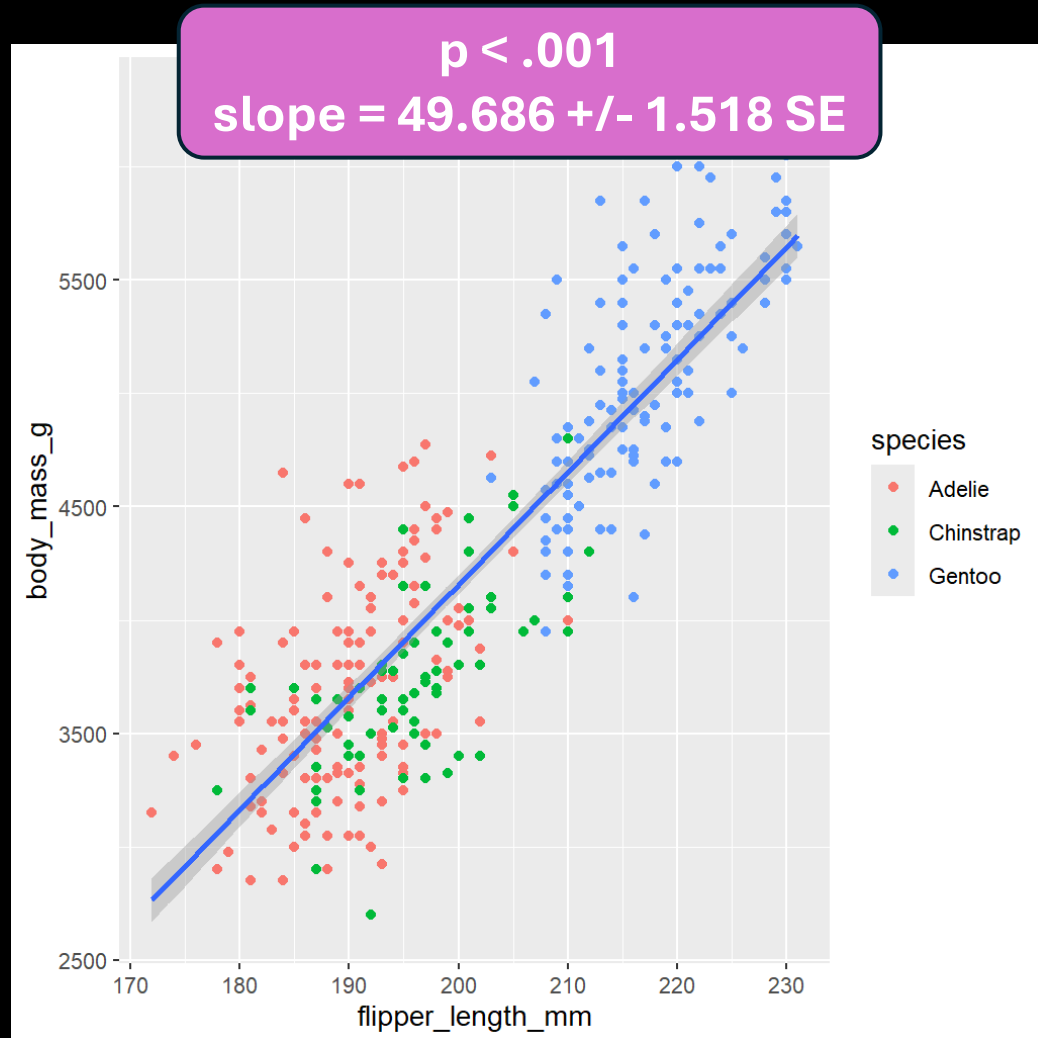
Part 2: Data Viz Continued



Part 2: Data Viz Continued

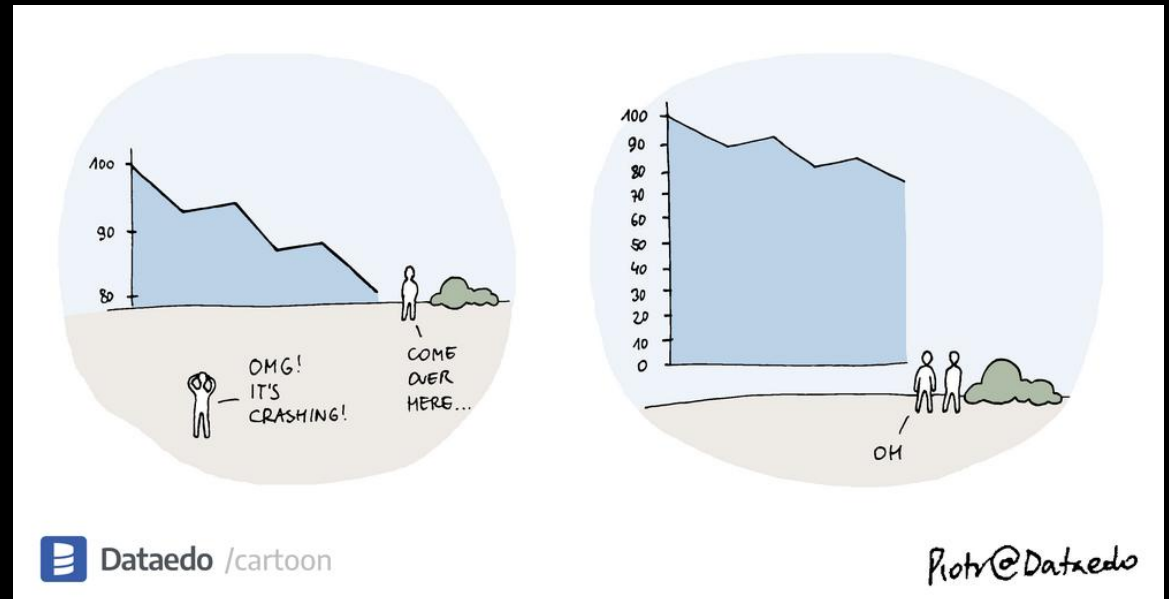


Part 2: Data Viz Continued



Other reasons data viz is important

- It can reveal previously unforeseen relationships
- Science communication
 - Most people are even more confused by stats than you are, and giving them a visual does wonders for communicating your results
- It's fun

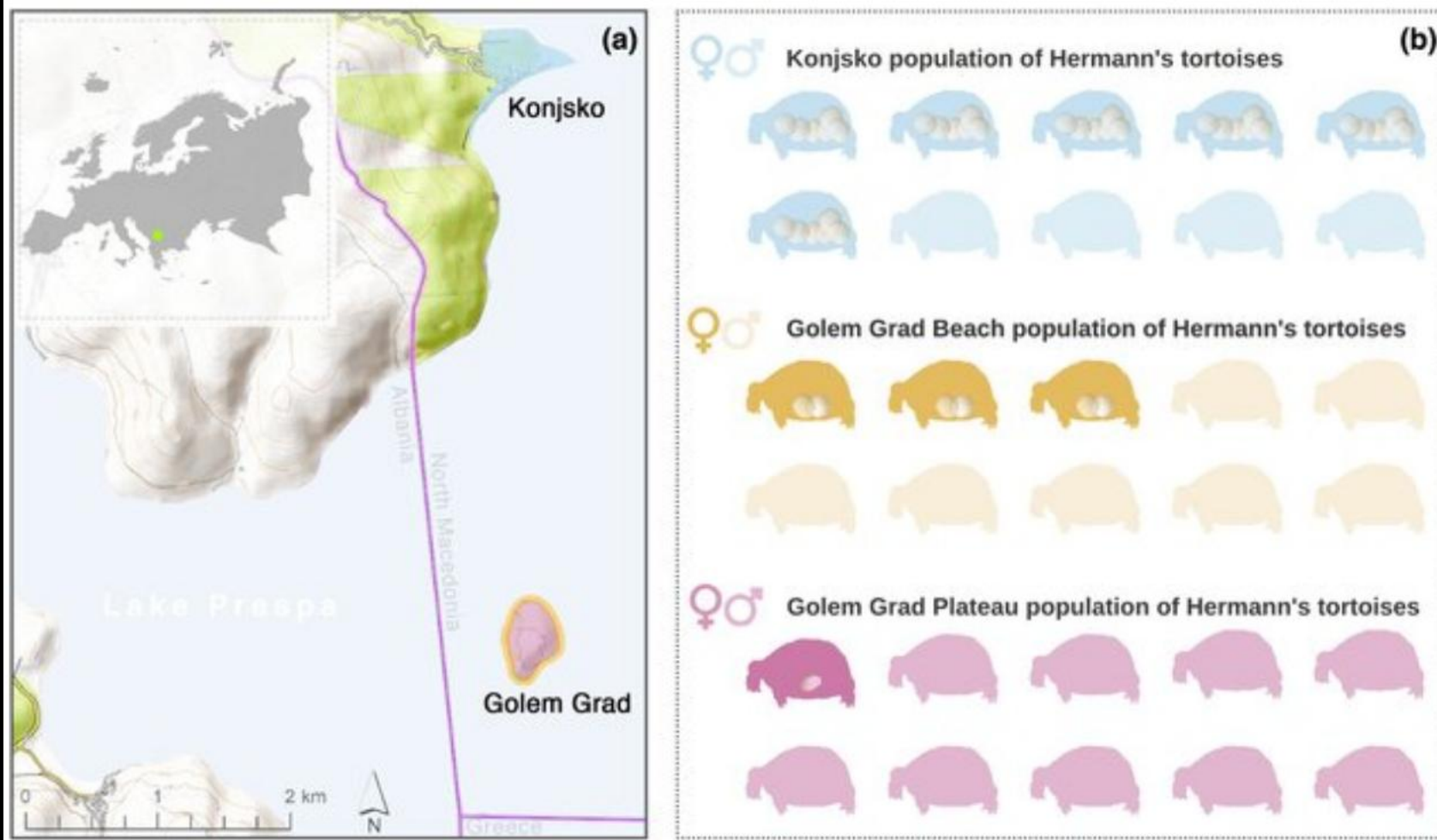


Let's make more visualizations!
(and learn more R along the way...)

The data we'll be looking at...

In an exceptionally dense island population of Hermann's tortoises in Lake Prespa in North Macedonia, sexually coercive males dramatically outnumber females, inflict severe copulatory injuries and put them at risk of fatal falls from the island plateau's sheer rock faces. Harassed females are emaciated, reproduce less frequently, produce smaller clutches and have lower annual survival rates compared to females from a neighbouring mainland population. Sixteen years of capture-recapture data reveal an ongoing extinction event and predict that the last island female will die in 2083.

Hermann's Tortoises



Hermann's Tortoises Data

`clutch_size_cleaned.csv`

variable	class	description
individual	integer	The ID of the individual.
age	integer	The age of the individual.
date	date	Date of the record.
locality	character	Locality.
eggs	integer	Number of eggs.
body_mass_grams	integer	The body mass of the individual in grams.
straight_carapace_length_mm	integer	The carapace length in millimeters.



Switch to ze code!



Additional Resources

- [Data visualization tips for clear communication | Geckoboard](#)
- [The R Graph Gallery – Help and inspiration for R charts](#)

R Vocabulary

- Working directory
 - `getwd()`, `setwd()`
- R Project

